

# Relations and Functions

- **Definition 1**

Let  $X$  be a set. A **relation**  $\sim$  on  $X$  is a rule that determines for every  $x, y \in X$  whether the statement

$$x \sim y$$

is true or false. If it is true, we say that  $x$  is related to  $y$ . If it is false, we write  $x \not\sim y$ .

- **Example 1.1**

Let  $X = \{1, 2, 3, 4\}$ , the following are examples of relations on  $X$ .

1. The relation  $\sim$  given by  $1 \sim 2, 2 \sim 3, 3 \sim 1, 4 \sim 4$ .
2. The relation  $\sim$  given by  $1 \sim 1, 2 \sim 2, 2 \sim 4$ .
3. The relation  $\sim$  given by  $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$ .

- **Definition 2**

Let  $X$  be a set and  $\sim$  a relation on  $X$ .

1. The relation is **reflexive** if  $x \sim x$  for every  $x \in X$ .
2. The relation is **symmetric** if for every  $x, y \in X$

$$x \sim y \implies y \sim x.$$

3. The relation is **antisymmetric** if for every  $x, y \in X$

$$x \sim y \wedge y \sim x \implies x = y.$$

4. The relation is **transitive** if for every  $x, y, z \in X$

$$x \sim y \wedge y \sim z \implies x \sim z.$$

- **Example 2.1**

On the set  $\mathbb{Z}$  consider the relation

$$x \sim y \iff x \leq y.$$

Note that

1.  $\sim$  is reflexive since  $x \leq x$  for every  $x \in \mathbb{Z}$ .
2.  $\sim$  is not symmetric since  $1 \leq 2$  but  $2 \not\leq 1$ .
3.  $\sim$  is antisymmetric because if  $x \leq y$  and  $y \leq x$  then  $x = y$ .
4.  $\sim$  is transitive since  $x \leq y$  and  $y \leq z$  implies that  $x \leq z$ .

- **Definition 3**

Let  $X$  be a set. An **equivalence relation** on  $X$  is a relation  $\sim$  that satisfies the following properties.

1.  $\sim$  is reflexive.
2.  $\sim$  is symmetric.
3.  $\sim$  is transitive.

- **Example 3.1**

Let  $X = \{1, 2, 3, 4\}$ . The following are examples of equivalence relations on  $X$ .

1. The relation  $\sim$  given by  $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$ .
2. The relation  $\sim$  given by  $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 2 \sim 3, 3 \sim 2, 1 \sim 4, 4 \sim 1$ .

- **Definition 4**

Let  $X$  be a set. An **order relation** on  $X$  is a relation  $\sim$  that satisfies the following properties.

1.  $\sim$  is reflexive.
2.  $\sim$  is antisymmetric.
3.  $\sim$  is transitive.

◦ **Example 4.1**

Let  $X = \{1, 2, 3, 4\}$ . The following are examples of order relations on  $X$ .

1. The relation  $\sim$  given by  $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$ .
2. The relation  $\sim$  given by  $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 1 \sim 2, 1 \sim 3, 1 \sim 4, 2 \sim 3, 2 \sim 4, 3 \sim 4$ .

• **Definition 5**

A **function**  $f: X \rightarrow Y$  is a correspondence of elements of  $X$  with elements of  $Y$  such that for all  $x \in X$  there exists a unique  $y \in Y$  such that  $y = f(x)$ .

$$\begin{aligned} f: X &\rightarrow Y \\ x &\mapsto y = f(x) \end{aligned}$$

The element  $y = f(x)$  is called the **image** of  $x$  under  $f$  and the element  $x$  is called the **preimage** of  $y$  under  $f$ .

• **Definition 6**

If  $f: X \rightarrow Y$  is a function, the **domain** of  $f$  denote  $\text{Dom}(f)$  is  $X$  and the **codomain** of  $\text{CoDom}(f)$  is  $Y$ . The **range** of  $f$ , denoted  $\text{Ran}(f)$ , is the set

$$\text{Ran}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$$

Note that  $\text{Ran}(f) \subseteq Y$ .

◦ **Example 6.1**

1. The function  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ .
2. The function  $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \sin(x)$ .
3. The function  $h: [0, \infty) \rightarrow \mathbb{R}, h(x) = \sqrt{x}$ .

◦ **Example 6.2**

Let  $X = \{1, 2, 3\}$  and  $Y = \{a, b, c\}$  the following examples of functions from  $X$  to  $Y$ .

1. The function  $f: X \rightarrow Y$  defined by

$$f(1) = a \quad f(2) = b \quad f(3) = c.$$

2. The function  $g: X \rightarrow Y$  defined by

$$g(1) = a \quad g(2) = a \quad g(3) = a.$$

3. The function  $h: X \rightarrow Y$  defined by

$$h(1) = a \quad h(2) = b \quad h(3) = a.$$

• **Definition 7**

Two functions  $f: X \rightarrow Y$  and  $g: Z \rightarrow W$  are **equal** if  $X = Z, Y = W$  and  $f(x) = g(x)$  for all  $x \in X$ .

◦ **Example 7.1**

1. Let  $X$  be a set. The **identity function** on  $X$  is the function  $\text{id}_x: X \rightarrow X$  given by  $\text{id}_x(x) = x$ .
2. Let  $X$  and  $Y$  be sets and  $c \in Y$ . The **constant function** with value  $c$  is the function  $K_c: X \rightarrow Y$  given by  $K_c(x) = c$ .
3. Let  $X$  be a set and  $A \subseteq X$ . The **inclusion function** of  $A$  in  $X$  is the function  $\iota: A \rightarrow X$  (or  $\iota: A \hookrightarrow X$ ) given by  $\forall x \in A, \iota(x) = x$ .

4. If  $f: X \rightarrow Y$  is a function and  $A \subseteq X$ , the **restriction** of  $f$  to  $A$  is the function  $f|_A: A \rightarrow Y$  given by  $f|_A(x) = f(x)$  for all  $x \in A$ .